

**САНКТ-ПЕТЕРБУРГСКИЙ
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
УНИВЕРСИТЕТ ИТМО**

Дисциплина: Бэк-энд разработка

**ОТЧЁТ
Лабораторная работа №4**

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Санкт-Петербург,

2026 г.

Задача

1. Выполнить предварительную настройку удалённого сервера и установить необходимые зависимости.
2. Развернуть существующее backend-приложение на удалённом сервере.
3. Настроить Nginx в качестве reverse проху для внешнего доступа к приложению.

Ход работы

1. Подготовка удалённого сервера

Для развёртывания использована виртуальная машина “recipes-lr4”, созданная в Yandex Cloud с Ubuntu 24.04.5 LTS. После подключения по SSH была выполнена установка обновлений системы и перезагрузка сервера. На сервер установлены Docker Engine, Docker Compose Plugin и Nginx. Docker настроен для работы обычного пользователя. Для внешнего HTTP-доступа в сетевой группе открыт TCP-порт 80, SSH-доступ осуществляется через TCP-порт 22.

Проверка Docker и Docker Compose на удалённом сервере:

```
lizaks@recipes-lr4:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
 linux-headers-6.8.0-60 linux-headers-6.8.0-60-generic linux-image-6.8.0-60-generic
 linux-modules-6.8.0-60-generic linux-tools-6.8.0-60 linux-tools-6.8.0-60-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
 docker-ce-rootless-extras pigz
Suggested packages:
 cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
 containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin
 pigz
0 upgraded, 7 newly installed, 0 to remove and 3 not upgraded.
Need to get 101 MB of archives.
After this operation, 383 MB of additional disk space will be used.
Get:1 http://mirror.yandex.ru/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.3.6-1~ubuntu.24.04~noble [23.2 MB]
Get:3 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:29.8.1-1~ubuntu.24.04~noble [17.6 MB]
Get:4 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:29.8.1-1~ubuntu.24.04~noble [24.3 MB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.37.1-1~ubuntu.24.04~noble [17.3 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:29.8.1-1~ubuntu.
```

```

macbook — lizaks@recipes-lr4: ~/ITMO-ACS-Backend-2026-C/БР2.2/Красноперова Елизавета/labs/lab3 — -ba...
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:29.8.1-1~ubuntu.
24.04~noble [10.2 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 5.5.1-1~ubuntu.24.04~n
oble [8012 kB]
Fetched 101 MB in 13s (7901 kB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 138191 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_2.3.6-1~ubuntu.24.04~noble_amd64.deb ...
Unpacking containerd.io (2.3.6-1~ubuntu.24.04~noble) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5%3a29.8.1-1~ubuntu.24.04~noble_amd64.deb ...
Unpacking docker-ce-cli (5:29.8.1-1~ubuntu.24.04~noble) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5%3a29.8.1-1~ubuntu.24.04~noble_amd64.deb ...
Unpacking docker-ce (5:29.8.1-1~ubuntu.24.04~noble) ...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.37.1-1~ubuntu.24.04~noble_amd64.deb ...
Unpacking docker-buildx-plugin (0.37.1-1~ubuntu.24.04~noble) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5%3a29.8.1-1~ubuntu.24.04~noble_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:29.8.1-1~ubuntu.24.04~noble) ...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../6-docker-compose-plugin_5.5.1-1~ubuntu.24.04~noble_amd64.deb ...
Unpacking docker-compose-plugin (5.5.1-1~ubuntu.24.04~noble) ...
Setting up docker-buildx-plugin (0.37.1-1~ubuntu.24.04~noble) ...
Setting up containerd.io (2.3.6-1~ubuntu.24.04~noble) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.
service.
Setting up docker-compose-plugin (5.5.1-1~ubuntu.24.04~noble) ...
Setting up docker-ce-cli (5:29.8.1-1~ubuntu.24.04~noble) ...
Setting up pigz (2.8-1) ...

```

```

macbook — lizaks@recipes-lr4: ~/ITMO-ACS-Backend-2026-C/БР2.2/Красноперова Елизавета/labs/lab3 — -ba...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.
service.
Setting up docker-compose-plugin (5.5.1-1~ubuntu.24.04~noble) ...
Setting up docker-ce-cli (5:29.8.1-1~ubuntu.24.04~noble) ...
Setting up pigz (2.8-1) ...
Setting up docker-ce-rootless-extras (5:29.8.1-1~ubuntu.24.04~noble) ...
Setting up docker-ce (5:29.8.1-1~ubuntu.24.04~noble) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lizaks@recipes-lr4:~$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-09-25 17:01:16 UTC; 3min 8s ago
 TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 1737 (dockerd)
      Tasks: 9
     Memory: 27.6M (peak: 29.5M)
        CPU: 517ms
    CGroup: /system.slice/docker.service
            └─1737 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

```

```

macbook — lizaks@recipes-lr4: ~/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова Елизавета/labs/lab3 — -ba...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lizaks@recipes-lr4:~$ sudo systemctl status docker --no-pager
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-09-25 17:01:16 UTC; 3min 8s ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 1737 (dockerd)
      Tasks: 9
     Memory: 27.6M (peak: 29.5M)
        CPU: 517ms
     CGroup: /system.slice/docker.service
             └─1737 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Sep 25 17:01:15 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:15.751020794Z" level=info msg="Restorin...start."
Sep 25 17:01:15 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:15.810416484Z" level=info msg="Deleting...atus 1"
Sep 25 17:01:15 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:15.818271305Z" level=info msg="Deleting...atus 1"
Sep 25 17:01:16 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:16.132656102Z" level=info msg="Loading ... done."
Sep 25 17:01:16 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:16.139159159Z" level=info msg="Docker d...=29.8.1
Sep 25 17:01:16 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:16.139276263Z" level=info msg="Initiali...ildkit"
Sep 25 17:01:16 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:16.394594069Z" level=info msg="Complete...zation"
Sep 25 17:01:16 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:16.403590394Z" level=info msg="Daemon h...zation"
Sep 25 17:01:16 recipes-lr4 dockerd[1737]: time="2026-09-25T17:01:16.403819840Z" level=info msg="API list...r.sock"
Sep 25 17:01:16 recipes-lr4 systemd[1]: Started docker.service - Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
lizaks@recipes-lr4:~$ sudo usermod -aG docker $USER

```

Установка и статус Nginx.:

```

macbook — lizaks@recipes-lr4: ~/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова Елизавета/labs/lab3 — -ba...

lizaks@recipes-lr4:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-60 linux-headers-6.8.0-60-generic linux-image-6.8.0-60-generic
  linux-modules-6.8.0-60-generic linux-tools-6.8.0-60 linux-tools-6.8.0-60-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 570 kB of archives.
After this operation, 1602 kB of additional disk space will be used.
Get:1 http://mirror.yandex.ru/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.18 [45.1 kB]
Get:2 http://mirror.yandex.ru/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.18 [525 kB]
Fetched 570 kB in 0s (30.0 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 138428 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.18_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.18) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.18_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.18) ...
Setting up nginx-common (1.24.0-2ubuntu7.18) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.18) ...
* Upgrading binary nginx [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...

```



```

macbook — lizaks@recipes-lr4: ~/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова Елизавета/labs/lab3 — -ba...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lizaks@recipes-lr4:~$ sudo systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-09-25 17:07:59 UTC; 35s ago
     Docs: man:nginx(8)
   Process: 2345 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 2347 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Main PID: 2379 (nginx)
    Tasks: 3 (limit: 4651)
   Memory: 2.2M (peak: 4.7M)
      CPU: 19ms
   CGroup: /system.slice/nginx.service
           └─2379 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2381 "nginx: worker process"
               └─2382 "nginx: worker process"

Sep 25 17:07:59 recipes-lr4 systemd[1]: Starting nginx.service - A high performance web server and a reverse...
Sep 25 17:07:59 recipes-lr4 systemd[1]: Started nginx.service - A high performance web server and a reverse...
Hint: Some lines were ellipsized, use -l to show in full.
lizaks@recipes-lr4:~$ git --version

```

2. Получение проекта на удалённом сервере

Исходный проект получен из репозитория GitHub ITMO-ACS-Backend-2026-С. Для развёртывания использована директория БР2.2/Красноперова Елизавета/labs/lab3. В ней находятся docker-compose.yml, Dockerfile отдельных сервисов и необходимые файлы конфигурации.

Клонирование репозитория и переход в директорию lab3:

```

lizaks@recipes-lr4:~$ git clone https://github.com/lizaveta/ITMO-ACS-Backend-2026-C.git
Cloning into 'ITMO-ACS-Backend-2026-C'...
remote: Enumerating objects: 464, done.
remote: Counting objects: 100% (188/188), done.
remote: Compressing objects: 100% (116/116), done.
remote: Total 464 (delta 82), reused 134 (delta 58), pack-reused 276 (from 3)
Receiving objects: 100% (464/464), 3.69 MiB | 4.89 MiB/s, done.
Resolving deltas: 100% (130/130), done.
lizaks@recipes-lr4:~$ cd ~/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова\ Елизавета/labs/lab3
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова Елизавета/labs/lab3$ pwd
/home/lizaks/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова Елизавета/labs/lab3
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова Елизавета/labs/lab3$ ls -la
total 408
drwxrwxr-x 4 lizaks lizaks 4096 Sep 25 17:30 .
drwxrwxr-x 5 lizaks lizaks 4096 Sep 25 17:30 ..
drwxrwxr-x 2 lizaks lizaks 4096 Sep 25 17:30 db
-rw-rw-r-- 1 lizaks lizaks 5030 Sep 25 17:30 docker-compose.yml
-rw-rw-r-- 1 lizaks lizaks 443 Sep 25 17:30 docker.env.example
-rw-rw-r-- 1 lizaks lizaks 96007 Sep 25 17:30 package-lock.json
-rw-rw-r-- 1 lizaks lizaks 584 Sep 25 17:30 package.json
drwxrwxr-x 6 lizaks lizaks 4096 Sep 25 17:30 services
-rw-rw-r-- 1 lizaks lizaks 285364 Sep 25 17:30 ЛР3_Красноперова_БР2.pdf
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БР2.2/Красноперова Елизавета/labs/lab3$ cat docker-compose.yml

```

3. Сборка контейнеров

Перед запуском приложения выполнена сборка Docker-образов командой `docker compose build`. Сборка успешно завершена для API Gateway, UserService, RecipeService и SocialService.

Успешное выполнение `docker compose build`..:

```
macbook — lizaks@recipes-lr4: ~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3 — -ba...
_PASSWORD||"311202",database:process.env.DB_NAME||"recipes_users",synchronize:true,logging:false,entities:[User]}});
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3$ docker compose build
[+] Building 89.1s (61/61) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 2.67kB 0.0s
=> [api-gateway internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 1.05kB 0.0s
=> [user-service internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 1.05kB 0.0s
=> [social-service internal] load build definition from Dockerfile 0.2s
=> => transferring dockerfile: 1.07kB 0.0s
=> [recipe-service internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 1.07kB 0.0s
=> [user-service internal] load metadata for docker.io/library/node:20-alpine 1.7s
=> [user-service internal] load .dockerignore 0.0s
=> => transferring context: 28 0.0s
=> [recipe-service builder 1/10] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6 4.8s
=> => resolve docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632 0.0s
=> => sha256:fff4e2c1b189bf87d63ad8bd07f7f4eb288d6f2b6a07a8bb44c60e8c075d2096 445B / 445B 0.2s
=> => sha256:b2cbbfe903b0821005780971ddc5892edcc4ce74c5a48d82e1d2b382edac3122 1.26MB / 1.26MB 0.6s
=> => sha256:4feea04c154301db6f4a496efa397b3db96603b1c009c797cfdde77bea8b3287 43.23MB / 43.23MB 3.3s
=> => sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB 0.9s
=> => extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 0.2s
=> => extracting sha256:4feea04c154301db6f4a496efa397b3db96603b1c009c797cfdde77bea8b3287 1.1s
=> => extracting sha256:b2cbbfe903b0821005780971ddc5892edcc4ce74c5a48d82e1d2b382edac3122 0.1s
=> => extracting sha256:fff4e2c1b189bf87d63ad8bd07f7f4eb288d6f2b6a07a8bb44c60e8c075d2096 0.0s
=> [social-service internal] load build context 0.2s
=> => transferring context: 155.99kB 0.0s
=> [recipe-service internal] load build context 0.2s
=> => transferring context: 169.49kB 0.0s
=> [user-service internal] load build context 0.1s
=> => transferring context: 126.83kB 0.0s
=> [api-gateway internal] load build context 0.1s
=> => transferring context: 115.12kB 0.0s
```



```
macbook — lizaks@recipes-lr4: ~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3 — -ba...
=> => exporting config sha256:424e88305635aea96043c04436a62673afaf3eeb6b6dcabf617b2e8900a47d10 0.1s
=> => exporting attestation manifest sha256:22ef76876928c13f0a5080aff38eb6b3ec6cdeace6613e64b2e47d 0.2s
=> => exporting manifest list sha256:978b161397ccc94b8e24dc4c9d591d185b0a7ddd66567f3df3c826ef91a54 0.1s
=> => naming to docker.io/library/recipes-microservices-lr3-social-service:latest 0.0s
=> => unpacking to docker.io/library/recipes-microservices-lr3-social-service:latest 7.2s
=> [user-service] resolving provenance for metadata file 10.5s
=> [social-service] resolving provenance for metadata file 9.8s
=> [recipe-service] resolving provenance for metadata file 8.9s
=> [api-gateway] resolving provenance for metadata file 1.5s
[+] build 4/4
✓ Image recipes-microservices-lr3-api-gateway Built 89.2s
✓ Image recipes-microservices-lr3-user-service Built 89.2s
✓ Image recipes-microservices-lr3-recipe-service Built 89.3s
✓ Image recipes-microservices-lr3-social-service Built 89.3s
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3$ docker compose up -d
[+] up 39/39
✓ Image rabbitmq:3.13-management-alpine Pulled 43.3s
✓ Image postgres:17-alpine Pulled 96.2s
✓ Network recipes-microservices-lr3_recipes-net Created 0.1s
✓ Volume recipes-microservices-lr3_rabbitmq_data Created 0.0s
✓ Volume recipes-microservices-lr3_recipe_db_data Created 0.1s
✓ Volume recipes-microservices-lr3_user_db_data Created 0.0s
✓ Volume recipes-microservices-lr3_social_db_data Created 0.1s
✓ Container lr3-rabbitmq Healthy 21.9s
✓ Container lr3-social-db Healthy 21.9s
✓ Container lr3-recipe-db Healthy 21.2s
✓ Container lr3-user-db Healthy 20.5s
✓ Container lr3-user-service Started 19.9s
✓ Container lr3-recipe-service Started 20.5s
✓ Container lr3-social-service Started 21.1s
✓ Container lr3-api-gateway Started 21.2s
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3$ docker compose ps
NAME STATUS IMAGE COMMAND SERVICE CREATED
```

4. Запуск приложения на удалённом сервере

Вся контейнерная система запущена командой `docker compose up -d`. Созданы общая Docker-сеть и именованные volumes для PostgreSQL и RabbitMQ. В систему входят API Gateway, UserService, RecipeService, SocialService, три PostgreSQL и RabbitMQ. После запуска базы данных и RabbitMQ получили состояние `healthy`, а приложения - состояние `Up`.

Вывод `docker compose ps` со всеми восемью контейнерами:

```
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3$ docker compose ps
NAME STATUS IMAGE PORTS COMMAND SERVICE CREATED
lr3-api-gateway Up 29 seconds recipes-microservices-lr3-api-gateway "docker-entrypoint.s..." api-gateway 50 seconds
ago Up 29 seconds 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
lr3-rabbitmq Up 50 seconds rabbitmq:3.13-management-alpine (healthy) "docker-entrypoint.s..." rabbitmq 51 seconds
ago Up 50 seconds 4369/tcp, 5671/tcp, 15671/tcp, 15691-15692/tcp, 25672/tcp, 0.0.0.0:5673->5672/tcp, [::]:5673->5672/tcp, 0.0.0.0:15673->15672/tcp, [::]:15673->15672/tcp
lr3-recipe-db Up 50 seconds postgres:17-alpine (healthy) "docker-entrypoint.s..." recipe-db 51 seconds
ago Up 50 seconds 5432/tcp
lr3-recipe-service Up 30 seconds recipes-microservices-lr3-recipe-service "docker-entrypoint.s..." recipe-service 51 seconds
ago Up 30 seconds 0.0.0.0:3002->3002/tcp, [::]:3002->3002/tcp
lr3-social-db Up 50 seconds postgres:17-alpine (healthy) "docker-entrypoint.s..." social-db 51 seconds
ago Up 50 seconds 5432/tcp
lr3-social-service Up 29 seconds recipes-microservices-lr3-social-service "docker-entrypoint.s..." social-service 50 seconds
ago Up 29 seconds 0.0.0.0:3003->3003/tcp, [::]:3003->3003/tcp
lr3-user-db Up 49 seconds postgres:17-alpine (healthy) "docker-entrypoint.s..." user-db 51 seconds
ago Up 49 seconds 5432/tcp
lr3-user-service Up 31 seconds recipes-microservices-lr3-user-service "docker-entrypoint.s..." user-service 51 seconds
ago Up 31 seconds 0.0.0.0:3001->3001/tcp, [::]:3001->3001/tcp
```

5. Проверка приложения внутри сервера

До настройки внешнего доступа выполнена проверка API Gateway на удалённом сервере. Запрос к `http://localhost:3000/api/recipes` вернул HTTP 200 OK и JSON-ответ `{"items":[],"total":0,"page":1,"limit":20}`. Это подтверждает работу API Gateway, RecipeService и PostgreSQL.

```
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3$ curl -i http://localhost:3000/api/recipes
HTTP/1.1 200 OK
X-Powered-By: Express
Content-Type: application/json; charset=utf-8
Content-Length: 42
ETag: W/"2a-69KgUQ3fmTKS5kydMDwFttW2HNU"
Date: Fri, 25 Sep 2026 17:46:47 GMT
Connection: keep-alive
Keep-Alive: timeout=5
{"items":[],"total":0,"page":1,"limit":20}lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/
```

6. Настройка Nginx reverse proxy

Для внешнего доступа настроен Nginx. Стандартный сайт отключён, создан отдельный файл `/etc/nginx/sites-available/recipes-lr4`. Nginx принимает HTTP-запросы на порту 80 и передаёт их API Gateway на `localhost:3000`. После подключения конфигурации выполнена проверка `nginx -t`, завершившаяся сообщениями «syntax is ok» и «test is successful». Затем конфигурация применена через `systemctl reload nginx`:

```
lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-С/БП2.2/Красноперова Елизавета/labs/lab3$ sudo nginx -T
```

```
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
```

```
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

```
# configuration file /etc/nginx/nginx.conf:
```

```
user www-data;
```

```
worker_processes auto;
```

```
pid /run/nginx.pid;
```

```
error_log /var/log/nginx/error.log;
```

```
include /etc/nginx/modules-enabled/*.conf;
```

```
events {
```

```
    worker_connections 768;
```

```
    # multi_accept on;
```



```
}
```

```
http {
```

```
    ##
```

```
    # Basic Settings
```

```
    ##
```

```
    sendfile on;
```

```
    tcp_nopush on;
```

```
    types_hash_max_size 2048;
```

```
    # server_tokens off;
```

```
    # server_names_hash_bucket_size 64;
```

```
    # server_name_in_redirect off;
```

```
    include /etc/nginx/mime.types;
```

```
    default_type application/octet-stream;
```

```
    ##
```

```
    # SSL Settings
```

```
    ##
```

```
    ssl_protocols TLSv1 TLSv1.1 TLSv1.2 TLSv1.3; # Dropping SSLv3, ref: POODLE
```

```
    ssl_prefer_server_ciphers on;
```

```
    ##
```

```
    # Logging Settings
```

```
    ##
```

```
    access_log /var/log/nginx/access.log;
```

```

##
# Gzip Settings
##

gzip on;

# gzip_vary on;
# gzip_proxied any;
# gzip_comp_level 6;
# gzip_buffers 16 8k;
# gzip_http_version 1.1;
# gzip_types text/plain text/css application/json application/javascript text/xml application/xml
application/xml+rss text/javascript;

##
# Virtual Host Configs
##

include /etc/nginx/conf.d/*.conf;
include /etc/nginx/sites-enabled/*;
}

#mail {
#   # See sample authentication script at:
#   # http://wiki.nginx.org/ImapAuthenticateWithApachePhpScript
#
#   # auth_http localhost/auth.php;
#   # pop3_capabilities "TOP" "USER";
#   # imap_capabilities "IMAP4rev1" "UIDPLUS";

```

```

#
#    server {
#        listen    localhost:110;
#        protocol  pop3;
#        proxy     on;
#    }
#
#    server {
#        listen    localhost:143;
#        protocol  imap;
#        proxy     on;
#    }
#}

```

configuration file /etc/nginx/mime.types:

```

types {
    text/html                html htm shtml;
    text/css                 css;
    text/xml                 xml;
    image/gif                gif;
    image/jpeg               jpeg jpg;
    application/javascript   js;
    application/atom+xml     atom;
    application/rss+xml      rss;

    text/mathml              mml;
    text/plain               txt;
    text/vnd.sun.j2me.app-descriptor jad;
    text/vnd.wap.wml         wml;
    text/x-component         htc;

```


image/avif	avif;
image/png	png;
image/svg+xml	svg svgz;
image/tiff	tif tiff;
image/vnd.wap.wbmp	wbmp;
image/webp	webp;
image/x-icon	ico;
image/x-jng	jng;
image/x-ms-bmp	bmp;
font/woff	woff;
font/woff2	woff2;
application/java-archive	jar war ear;
application/json	json;
application/mac-binhex40	hqx;
application/msword	doc;
application/pdf	pdf;
application/postscript	ps eps ai;
application/rtf	rtf;
application/vnd.apple.mpegurl	m3u8;
application/vnd.google-earth.kml+xml	kml;
application/vnd.google-earth.kmz	kmz;
application/vnd.ms-excel	xls;
application/vnd.ms-fontobject	eot;
application/vnd.ms-powerpoint	ppt;
application/vnd.oasis.opendocument.graphics	odg;
application/vnd.oasis.opendocument.presentation	odp;
application/vnd.oasis.opendocument.spreadsheet	ods;
application/vnd.oasis.opendocument.text	odt;

application/vnd.openxmlformats-officedocument.presentationml.presentation

pptx;

application/vnd.openxmlformats-officedocument.spreadsheetml.sheet

xlsx;

application/vnd.openxmlformats-officedocument.wordprocessingml.document

docx;

application/vnd.wap.wmlc wmlc;

application/wasm wasm;

application/x-7z-compressed 7z;

application/x-cocoa cco;

application/x-java-archive-diff jardiff;

application/x-java-jnlp-file jnlp;

application/x-makeself run;

application/x-perl pl pm;

application/x-pilot prc pdb;

application/x-rar-compressed rar;

application/x-redhat-package-manager rpm;

application/x-sea sea;

application/x-shockwave-flash swf;

application/x-stuffit sit;

application/x-tcl tcl tk;

application/x-x509-ca-cert der pem crt;

application/x-xpinstall xpi;

application/xhtml+xml xhtml;

application/xspf+xml xspf;

application/zip zip;

application/octet-stream bin exe dll;

application/octet-stream deb;

application/octet-stream dmg;

application/octet-stream iso img;

```

application/octet-stream      msi msp msm;

audio/midi                    mid midi kar;
audio/mpeg                     mp3;
audio/ogg                      ogg;
audio/x-m4a                    m4a;
audio/x-realaudio              ra;


video/3gpp                     3gpp 3gp;
video/mp2t                     ts;
video/mp4                      mp4;
video/mpeg                     mpeg mpg;
video/ogg                      ogv;
video/quicktime                mov;
video/webm                     webm;
video/x-flv                    flv;
video/x-m4v                    m4v;
video/x-matroska               mkv;
video/x-mng                    mng;
video/x-ms-asf                 asx asf;
video/x-ms-wmv                 wmv;
video/x-msvideo                avi;
}

```

configuration file /etc/nginx/sites-enabled/default:

##

You should look at the following URL's in order to grasp a solid understanding

of Nginx configuration files in order to fully unleash the power of Nginx.

<https://www.nginx.com/resources/wiki/start/>

https://www.nginx.com/resources/wiki/start/topics/tutorials/config_pitfalls/

<https://wiki.debian.org/Nginx/DirectoryStructure>

```
#  
# In most cases, administrators will remove this file from sites-enabled/ and  
# leave it as reference inside of sites-available where it will continue to be  
# updated by the nginx packaging team.  
#  
# This file will automatically load configuration files provided by other  
# applications, such as Drupal or Wordpress. These applications will be made  
# available underneath a path with that package name, such as /drupal8.  
#  
# Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.  
##
```

```
# Default server configuration
```

```
#
```

```
server {
```

```
    listen 80 default_server;
```

```
    listen [::]:80 default_server;
```

```
    # SSL configuration
```

```
    #
```

```
    # listen 443 ssl default_server;
```

```
    # listen [::]:443 ssl default_server;
```

```
    #
```

```
    # Note: You should disable gzip for SSL traffic.
```

```
    # See: https://bugs.debian.org/773332
```

```
    #
```

```
    # Read up on ssl_ciphers to ensure a secure configuration.
```

```
    # See: https://bugs.debian.org/765782
```

```
    #
```

```
    # Self signed certs generated by the ssl-cert package
```

```
    # Don't use them in a production server!
```

```
#
# include snippets/snakeoil.conf;

root /var/www/html;

# Add index.php to the list if you are using PHP
index index.html index.htm index.nginx-debian.html;

server_name _;

location / {
    # First attempt to serve request as file, then
    # as directory, then fall back to displaying a 404.
    try_files $uri $uri/ =404;
}

# pass PHP scripts to FastCGI server
#
#location ~ /\.php$ {
#    include snippets/fastcgi-php.conf;
#
#    # With php-fpm (or other unix sockets):
#    fastcgi_pass unix:/run/php/php7.4-fpm.sock;
#    # With php-cgi (or other tcp sockets):
#    fastcgi_pass 127.0.0.1:9000;
#}

# deny access to .htaccess files, if Apache's document root
# concurs with nginx's one
#
#location ~ /\.ht {
```



```

        #    deny all;
        #}
    }

# Virtual Host configuration for example.com
#
# You can move that to a different file under sites-available/ and symlink that
# to sites-enabled/ to enable it.
#
#server {
#    listen 80;
#    listen [::]:80;
#
#    server_name example.com;
#
#    root /var/www/example.com;
#    index index.html;
#
#    location / {
#        try_files $uri $uri/ =404;
#    }
#}

```

7. Проверка reverse proxy

Запрос к `http://localhost/api/recipes` вернул HTTP 200 OK, заголовок `Server: nginx/1.24.0 (Ubuntu)` и JSON-ответ приложения. Это подтверждает прохождение запроса через Nginx к API Gateway.

```
ks@recipes-lr4:~/ITMO-ACS-Backend-2026-C/БП2.2/Красноперова Елизавета/labs/lab3$ curl -i http://localhost/api/recipes
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Fri, 25 Sep 2026 17:51:30 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 42
Connection: keep-alive
X-Powered-By: Express
ETag: W/"2a-69KgUQ3fmTKS5kydMDwFttW2HNU"

{"items":[],"total":0,"page":1,"limit":20}lizaks@recipes-lr4:~/ITMO-ACS-Backend-2026-C/БП2.2/Красноперова Елизавета/
```

8. Проверка внешнего доступа

С локального компьютера выполнен запрос к публичному IP виртуальной машины: `http://51.250.107.202/api/recipes`. Получен HTTP 200 OK и корректный JSON-ответ. Проверена цепочка «клиент - Yandex Cloud - Nginx - API Gateway - микросервисы».

```
MacBook-Air:~ macbook$ curl -i http://51.250.107.202/api/recipes
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Fri, 25 Sep 2026 17:52:32 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 42
Connection: keep-alive
X-Powered-By: Express
ETag: W/"2a-69KgUQ3fmTKS5kydMDwFttW2HNU"

{"items":[],"total":0,"page":1,"limit":20}MacBook-Air:~ macbook$ curl -i -X POST http://51.250.107.202/api/auth/register \
```

9. Функциональная проверка через внешний адрес

Через публичный адрес выполнена регистрация тестового пользователя: `POST /api/auth/register` вернул HTTP 201 Created и JWT. Затем `GET /api/users/me` с этим JWT вернул HTTP 200 OK и данные пользователя. Проверена работа маршрутизации Gateway, UserService, PostgreSQL и JWT-аутентификации через Nginx.

```
{"items":[],"total":0,"page":1,"limit":20}MacBook-Air:~ macbook$ curl -i -X POST http://51.250.107.202/api/auth/register \
> -H "Content-Type: application/json" \
> -d '{"username":"lr4_test","email":"lr4_test@example.com","password":"Test12345!"}'
HTTP/1.1 201 Created
Server: nginx/1.24.0 (Ubuntu)
Date: Fri, 25 Sep 2026 18:33:41 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 227
Connection: keep-alive
X-Powered-By: Express
ETag: W/"e3-Xe+JUt2SGLDaa6GpwNKQVvMBUFw"

{"token":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJlhdCI6MTc5MDM2MTIyMSwiZXhwIjoxNzkwOTY2MDI5fQ.EBTJij-2q4vcd7nkJBe4AEazogAk33yTq4KU5o9CJu4","user":{"user_id":1,"username":"lr4_test","avatar_url":null,"bio":null}}MacBook-Air:~ macbook$ curl -i http://51.250.107.202/api/users/me \
```

```
{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJlhdCI6MTc5MDM2MTIyMSwiZXhwIjoxNzkwOTY2MDIxfQ.EBTJij-2q4vcd7nkJBe4AEazogAk33yTq4KU5o9CJu4", "user": { "user_id": 1, "username": "lr4_test", "avatar_url": null, "bio": null } } MacBook-Air:~ macbook$ curl -i http://51.250.107.202/api/users/me \
[> -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJlhdCI6MTc5MDM2MTIyMSwiZXhwIjoxNzkwOTY2MDIxfQ.EBTJij-2q4vcd7nkJBe4AEazogAk33yTq4KU5o9CJu4"
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Fri, 25 Sep 2026 18:34:58 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 149
Connection: keep-alive
X-Powered-By: Express
ETag: W/"95-yErZ6+LzgDc5b91DjUlfQF85JH0"

{ "user_id": 1, "username": "lr4_test", "email": "lr4_test@example.com", "avatar_url": null, "bio": null, "created_at": "2026-09-25T18:33:41.119Z", "role": "user" } MacBook-Air:~ macbook$
```

Вывод

В ходе лабораторной работы выполнено развёртывание существующего микросервисного backend-приложения на удалённой виртуальной машине в Yandex Cloud. На сервере установлены Docker, Docker Compose и Nginx. Приложение собрано и запущено в контейнерах: API Gateway, три микросервиса, три базы PostgreSQL и RabbitMQ.

Для внешнего доступа настроен Nginx в качестве reverse proxy. Конфигурация проверена с помощью `nginx -t`, а работоспособность подтверждена запросами с сервера и с локального компьютера через публичный IP. Дополнительная проверка регистрации и JWT-аутентификации показала корректную работу приложения через внешний HTTP-доступ.